KCH-MedRank: An Interpretable Medical Evidence Control Layer for Evidence-Grounded Retrieval-Augmented Generation

Anonymous Authors

May 22, 2026

Abstract

Medical retrieval-augmented generation systems require evidence selection that is accurate, medically grounded, and inspectable before answer generation. This paper presents KCH-MedRank, an interpretable medical evidence control layer implemented as a reproducible knowledge-constrained local hypergraph learning-to-rank framework for biomedical evidence retrieval and reranking. Enhanced candidate generation improves the top-100 evidence ceiling, and KCH-MedRank improves top-10 evidence retrieval over strong Hybrid RRF on the held-out BioASQ test split. Against a true MedCPT Cross-Encoder baseline on the same candidate pool, KCH-MedRank significantly improves Recall@10, while the rank-sensitive MRR@10 and nDCG@10 gains remain positive but non-significant. A controlled reranking-stage timing comparison shows that KCH-MedRank obtains these metrics while running 19.3× faster than MedCPT Cross-Encoder on the held-out test pool. A diagnostic PubMedQA Qwen3-8B pilot suggests that improved evidence selection can affect answer accuracy and macro-F1 in a controlled downstream loop, while citation support is already high for all compared evidence sources. The results position KCH-MedRank as a white-box evidence selection module for medical RAG rather than as a clinical decision system.

1 Introduction

Evidence-grounded medical question answering requires systems to retrieve relevant biomedical evidence before producing or selecting an answer. Retrieval-augmented generation (RAG) provides a general framework for combining retrieved evidence with language models [8], and recent medical benchmarking work has shown that retrieval choices strongly affect medical QA performance [15]. However, standard lexical or dense retrieval often ranks passages by pairwise similarity only. Biomedical evidence is frequently organized through higher-order relations among questions, passages, MeSH concepts, biomedical entities, documents, and relation sources.

This paper studies a practical, domain-specific evidence control layer rather than a new generic RAG paradigm. We present KCH-MedRank, a knowledge-constrained local hypergraph learning-to-rank framework that acts as a white-box evidence selection module for biomedical RAG. The method keeps candidate generation reproducible with BM25, dense retrieval, and Reciprocal Rank Fusion [2, 11], then builds a local cross-granularity hypergraph around each question and its top candidate passages.

The contribution is a focused and interpretable evidence reranking framework. KCH-MedRank uses biomedical semantic reranking scores, MeSH hierarchy-aware features, local hypergraph diffusion and centrality, and query-group LambdaMART ranking to improve evidence retrieval and expose the evidence signals used by the reranker. Hypergraph signals are used as interpretable ranking features rather than as a hand-tuned standalone scoring formula. The system does not claim to replace clinical judgment, provide clinical diagnosis, or guarantee answer correctness.

The experiments use public biomedical QA resources and emphasize evidence retrieval metrics, ablations, and controlled supporting analyses. On the BioASQ evidence retrieval setting, enhanced candidate generation raises the held-out top-100 recall ceiling, and KCH-MedRank improves top-10 evidence retrieval over original Hybrid RRF and a true MedCPT Cross-Encoder baseline. PubMedQA is used as a secondary diagnostic setting: retrieval experiments test evidence coverage transfer, and a fixed-prompt Qwen3-8B pilot examines whether evidence selection changes downstream answer accuracy and support behavior.

The contributions are:

- a reproducible two-stage biomedical evidence retrieval pipeline with BM25, dense retrieval, Hybrid RRF, enhanced candidate fusion, and held-out evaluation;
- a local knowledge-constrained hypergraph representation connecting question, passage, document, entity, MeSH concept, hierarchy, and auxiliary relation signals;
- a learning-to-rank reranker that combines retrieval, biomedical semantic, MeSH hierarchy, entity, relation, and hypergraph features under query-level groups;
- an evidence-oriented evaluation including Recall@k, Hit@k, MRR@k, nDCG@k, hard reranking subsets, paired bootstrap significance tests, ablations, failure analysis, and a controlled PubMedQA Qwen3-8B generation pilot.

The failure analysis also shows lost gold passages, so the paper treats the method as a net retrieval improvement with clear remaining failure modes rather than as a complete solution for clinical medical QA.

2 Related Work

2.1 Medical Retrieval-Augmented Question Answering

Medical RAG benchmarking has recently made systematic evaluation of corpora, retrievers, and backbone models more visible. The MedRAG/MIRAGE benchmark evaluates medical RAG configurations over multiple QA datasets and highlights that retrieval settings have a large effect on downstream medical QA [15]. Iterative medical RAG extends this direction by generating follow-up retrieval queries for complex questions [16]. This work is narrower: it focuses on evidence retrieval and reranking before making strong generation claims.

2.2 Biomedical QA Datasets and Biomedical Retrieval

BioASQ provides a long-running biomedical semantic indexing and QA benchmark [12], and BioASQ-QA offers manually curated questions, reference answers, documents, snippets, and concepts for biomedical QA research [7]. PubMedQA is a complementary biomedical research question answering dataset with yes/no/maybe labels over PubMed abstracts [4]. For biomedical semantic retrieval, MedCPT uses large-scale PubMed search logs to train a biomedical retriever and reranker [5].

2.3 Graph and Hypergraph RAG

GraphRAG-style methods organize retrieved information using graph structure, with local-to-global search proposed for query-focused summarization [3]. Medical Graph RAG connects user documents, credible medical sources, and controlled vocabularies for medical response grounding [14]. HyperGraphRAG argues that hyperedges can represent n-ary relations beyond ordinary pairwise graph edges [9], and cross-granularity HGRAG models entities and passages at different retrieval granularities [13]. A recent Medical HyperRAG preprint studies patient-centered clinical QA with a ClinBridge hypergraph over records, cases, and guidelines [17]. KCH-MedRank follows the high-order relation motivation, but differs in scope: it uses public biomedical QA resources, avoids private clinical records, and evaluates a local evidence-reranking module rather than a full patient-centered clinical decision-support framework.

2.4 Controlled Biomedical Knowledge

MeSH is a controlled, hierarchically organized vocabulary maintained by the National Library of Medicine for indexing and searching biomedical information [10]. PrimeKG integrates multimodal precision-medicine relations across diseases, proteins, drugs, phenotypes, and other biomedical entities [1]. In this study, MeSH hierarchy features are central to the medical constraint design, while PrimeKG is treated as an auxiliary relation source because exact-name coverage is sparse.

3 Task Definition

Given a medical question q, a biomedical corpus C, and optional knowledge sources K, the system returns a ranked evidence list E and optionally an answer a. The primary task is evidence retrieval and reranking; answer generation is secondary.

For each question, a first-stage retriever returns a candidate pool $D_q = \{d_1, \ldots, d_M\}$ from C. A reranker then estimates an improved ordering over D_q using retrieval scores, biomedical semantic scores, concept and entity features, and local hypergraph features. Evaluation emphasizes Recall@k, MRR@k, nDCG@k, and evidence coverage.

The formal output is:

$$(a, E, P),$$

where a is optional in the evidence-focused setting, E is the ranked supporting evidence list, and P is an optional interpretable entity, concept, or hypergraph path used for case analysis.

4 Method

4.1 First-Stage Retrieval

The first stage retrieves a high-recall candidate pool before any knowledge-aware reranking is applied. We use BM25 as the lexical baseline, dense retrieval as the semantic baseline, and Reciprocal Rank Fusion (RRF) as the default hybrid retriever. For a retrieval source s, let $r_s(q, d)$ be the rank of passage d for question q. The fused score is:

$$\mathrm{RRF}(q, d) = \sum_{s \in S} \frac{w_s}{k + r_s(q, d)},$$

where S is the set of retrieval sources, w_s is a source weight, and $k = 60$ in the reported experiments. The original Hybrid RRF uses BM25 and dense retrieval with equal weights.

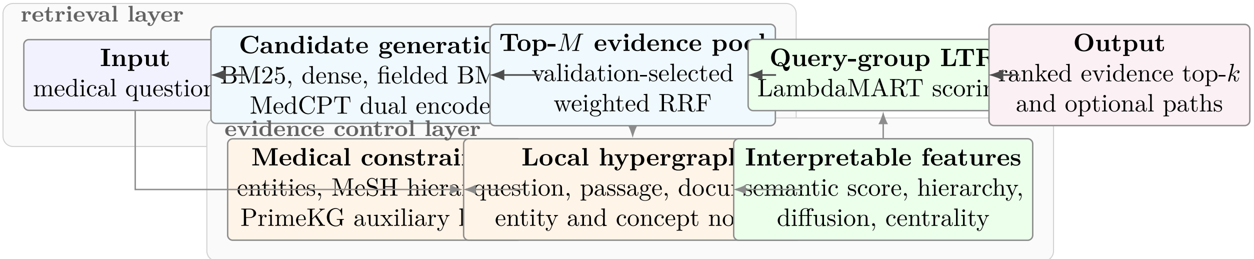

Figure 1: Publication-style overview of KCH-MedRank. Candidate generation provides a high-recall evidence pool; the evidence-control layer converts biomedical constraints and local hypergraph structure into interpretable features; the query-group ranker produces the final evidence order for downstream medical RAG.

The enhanced candidate-generation path is designed to raise the top-100 ceiling for reranking. It adds fielded BM25 with title, text, and MeSH expansion fields, plus MedCPT dual-encoder dense retrieval as an additional biomedical semantic source. Fusion weights are selected on the validation split only. The selected w122 configuration gives BM25 and dense retrieval weight 1.0, fielded BM25 weight 1.2, and MedCPT dual-encoder retrieval weight 0.2. This setting is selected for validation Recall@100 rather than test performance, because reranking can improve only passages already present in the candidate pool.

4.2 Knowledge-Constrained Local Hypergraph

For each question and its top-M candidates, KCH-MedRank builds a local hypergraph $H_q = (V_q, \mathcal{E}_q)$. The node set includes a question node, candidate passage nodes, document nodes, biomedical entity nodes, MeSH concept nodes, optional MeSH tree nodes, and optional relation nodes. The hyperedge set connects multiple nodes at once and therefore represents higher-order relations that are not naturally captured by pairwise similarity alone.

The main hyperedge types are question-entity, question-MeSH, question-passage candidate, passage-entity, passage-document, document-MeSH, shared-entity passages, entity-type concept, MeSH hierarchy, and auxiliary PrimeKG relation edges. Pairwise graph and no-medical-knowledge variants are constructed from the same candidate pool for ablation: the pairwise variant decomposes multi-node hyperedges into anchor-based pairwise edges, while the no-knowledge variant keeps only rank-band and candidate structure.

The local design avoids constructing a global graph over millions of biomedical passages. It also keeps the method interpretable: feature inspection and case studies can report which concept overlaps, hierarchy links, shared candidate clusters, or hypergraph centrality signals influenced reranking.

4.3 Hypergraph Diffusion and Structural Features

KCH-MedRank uses hypergraph diffusion as an interpretable feature rather than as a standalone ranking formula. The seed distribution places mass on the question node, question entity nodes, and question MeSH nodes. At each iteration, scores are propagated through weighted hyperedges:

$$x^{(t+1)} = (1-\alpha)x^{(0)} + \alpha \cdot \mathrm{Propagate}(x^{(t)}, \mathcal{E}_q),$$

where $\alpha = 0.85$ and the implementation uses three diffusion iterations. Passage-level diffusion scores are then min-max normalized within each question candidate set. We also compute weighted

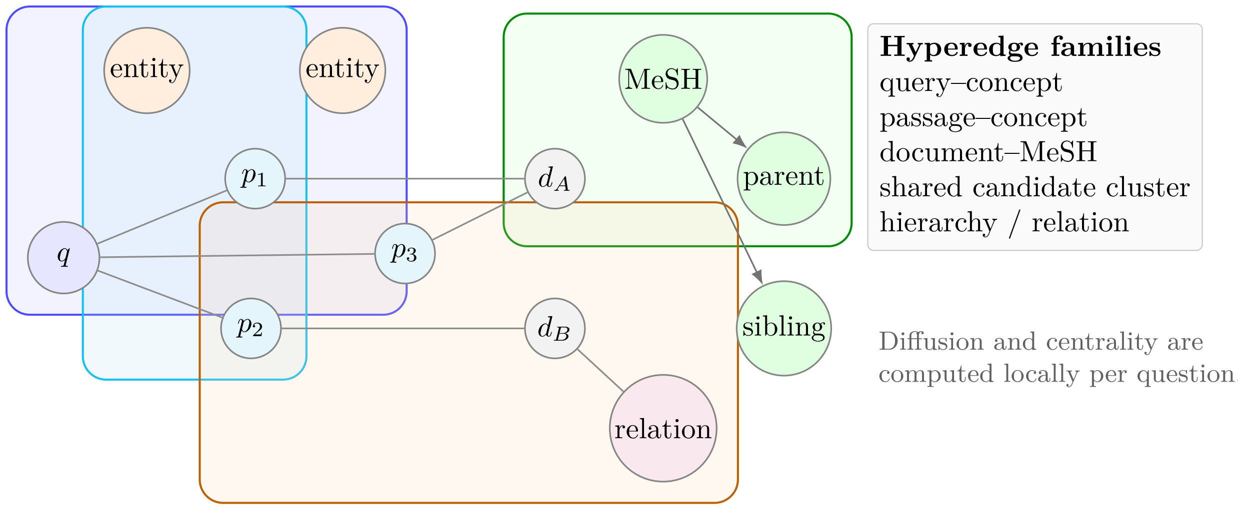

Figure 2: Local knowledge-constrained hypergraph used by KCH-MedRank. Hyperedges connect more than two typed nodes, allowing candidate passages to receive evidence from shared biomedical entities, MeSH hierarchy structure, document annotations, and auxiliary relation signals.

degree centrality, local node and hyperedge counts, shared-entity edge counts, document-MeSH edge counts, and optional hierarchy-edge counts.

4.4 Biomedical Knowledge Features

The reranker uses several medically motivated feature groups. Entity features include entity overlap count, entity Jaccard similarity, question entity coverage, and passage entity count. Exact MeSH features include MeSH overlap, MeSH Jaccard similarity, question MeSH coverage, and passage MeSH count. Hierarchy-aware MeSH features include exact descriptor overlap, parent match, ancestor match, sibling match, tree-distance similarity, query concept coverage, passage hierarchy coverage, passage concept specificity, and shared candidate clusters over MeSH terms or parent concepts.

PrimeKG-derived features are included only as auxiliary relation signals: relation count, question relation coverage, and local relation-edge count. Because exact-name relation coverage is sparse, these features are not treated as the main source of improvement.

4.5 Learning-to-Rank

The main supervised reranker is a query-group learning-to-rank model: a LightGBM gradient boosted tree ranker trained with a LambdaMART-style objective [6]. Each question forms one ranking group, candidate passages form rows, and binary relevance labels are derived from gold evidence annotations. The feature vector combines retrieval scores, biomedical semantic scores, entity features, exact and hierarchy-aware MeSH features, auxiliary relation features, and local hypergraph structural features.

Hyperparameters are selected on the validation split. The reported search space is number of leaves $\{15, 31\}$, learning rate $\{0.03, 0.05\}$, estimators $\{80, 160\}$, and blend weight $\{0, 0.1, 0.2, 0.35\}$, where the blend weight interpolates model scores with the original candidate rank score. Selection prioritizes validation MRR@10, then Recall@10 and nDCG@10. After validation selection, the final model is trained on train plus validation questions and evaluated once on the held-out test split.

Algorithm 1: KCH-MedRank local evidence control and reranking

1. **Candidate retrieval.** For question q, retrieve top-M passages from BM25, dense retrieval, fielded BM25, and MedCPT dual-encoder sources; fuse them with validation-selected weighted RRF.

2. **Local hypergraph construction.** Create nodes for q, candidate passages, documents, extracted biomedical entities, MeSH descriptors, MeSH hierarchy concepts, and optional relation endpoints. Add hyperedges for query-concept, passage-concept, document-MeSH, shared-concept candidate clusters, hierarchy links, candidate membership, and auxiliary PrimeKG relations.

3. **Hyperedge weighting and normalization.** Assign edge weights by source reliability and local evidence strength: candidate membership edges use normalized RRF scores; entity and MeSH edges use normalized overlap/coverage; hierarchy edges use inverse tree distance; relation edges are down-weighted because of sparse exact-name coverage. Within each local graph, incident edge weights are normalized before propagation.

4. **Diffusion propagation.** Initialize mass on the question, query-entity, and query-MeSH nodes. Run three restart diffusion iterations with $\alpha = 0.85$, then normalize passage diffusion scores within the candidate set.

5. **Passage feature extraction.** For each passage, concatenate retrieval scores and ranks, biomedical semantic scores, entity overlap, MeSH exact and hierarchy features, auxiliary relation features, diffusion scores, centrality, and local graph-count features.

6. **Query-group LambdaMART scoring.** Train a LightGBM LambdaMART ranker with one group per question on train plus validation after validation-only model selection. Score test candidates once and sort passages by the learned score to produce the final evidence ranking.

Figure 3: Compact pseudocode for local hypergraph construction, normalization, diffusion, feature extraction, and query-group learning-to-rank.

Feature importance in Table 1 shows that semantic and first-stage retrieval signals remain central, while MeSH cluster, passage specificity, and hypergraph centrality features contribute interpretable biomedical and structural evidence. This supports a cautious claim: KCH-MedRank improves reranking by combining retrieval strength, biomedical semantics, and local knowledge-constrained structure, not by relying on one isolated knowledge source.

5 Experimental Setup

5.1 Dataset

The primary dataset is `rag-datasets/rag-mini-bioasq`, used as a compact BioASQ-style biomedical evidence retrieval setting. The full processed set contains 4,719 questions. We use a deterministic question-level split based on question identifier buckets: training questions are all buckets except validation and test, validation uses bucket remainder 3, and test uses bucket remainder 4 under modulo 5. In the enhanced BioASQ experiment, this gives 2,832 training questions, 944

Table 1: Top KCH-MedRank feature importances from the enhanced BioASQ experiment.

Feature	Importance
Biomedical semantic score	505
Hybrid score	416
Dense score	376
BM25 rank score	349
Biomedical semantic rank score	304
Passage entity count	246
Dense rank score	242
BM25 score	215
Shared MeSH term cluster size	197
Shared MeSH parent cluster size	174
Hypergraph degree centrality	149
Passage MeSH specificity	135
Hypergraph diffusion score	130

validation questions, and 943 held-out test questions. The split is fixed before model selection.

Each question is paired with a set of gold evidence passage identifiers. Evaluation is performed at the passage level. The reranker receives only candidates returned by the first-stage retrieval pool, so Recall@100 of the candidate generator is reported as an upper-bound analysis. The split is question-level, and passages from test questions are not used as positive training labels for the reranker. PubMedQA is used as a secondary diagnostic dataset rather than the main result source.

5.2 Baselines

We compare against both first-stage retrieval baselines and reranking baselines. The first-stage baselines are BM25, dense biomedical retrieval, original Hybrid BM25 + dense RRF, fielded BM25 with MeSH expansion, MedCPT dual-encoder retrieval, and enhanced Hybrid RRF. Enhanced Hybrid RRF is not tuned on the test split; its selected w122 weights are chosen by validation Recall@100.

The reranking baselines are MedCPT Cross-Encoder on the same enhanced candidate pool, retrieval-feature-only LambdaMART, LambdaMART with biomedical semantic features but without hypergraph features, pairwise graph learning-to-rank, hypergraph learning-to-rank without medical knowledge, and full KCH-MedRank. We also report leave-one-feature-group ablations that remove the biomedical semantic score, MeSH hierarchy features, biomedical entity features, hypergraph diffusion and centrality, and PrimeKG relation features.

5.3 Evaluation

The main retrieval metrics are Recall@k, Hit@k, MRR@k, nDCG@k, and evidence coverage for $k \in \{1, 3, 5, 10, 20, 50, 100\}$. With multiple gold passages per question, Recall@k is macro evidence coverage: for each question, it is the fraction of that question's gold passages appearing in the top k, then averaged over questions. Hit@k is query-level success: it is 1 if at least one gold passage appears in the top k and 0 otherwise. MRR@k is also query-level: it is the reciprocal rank of the first gold passage within the top k, or 0 if no gold passage appears. nDCG@k uses qrel relevance values as gains; because the current evidence labels are binary, it reduces to binary-gain DCG normalized by the ideal ordering of all available gold passages up to k. Evidence coverage is

Table 2: Validation-selected enhanced candidate generation. Fusion weights are selected only on the validation split.

Fusion run	BM25	Dense	Fielded BM25	MedCPT DE	Val. R@100	Test R@100
Original Hybrid	1.0	1.0	0.0	0.0	0.6325	0.6077
w052	1.0	1.0	0.5	0.2	0.6391	0.6163
w084	1.0	1.0	0.8	0.4	0.7329	0.7084
w102	1.0	1.0	1.0	0.2	0.7529	0.7285
w122, selected	1.0	1.0	1.2	0.2	0.7568	0.7388

reported as the same macro passage recall value as Recall@k to connect the metric to the medical evidence-grounding interpretation.

Statistical testing uses paired bootstrap over held-out test questions with 10,000 resamples and a fixed random seed. We report confidence intervals and two-sided p-values against original or enhanced Hybrid RRF and against the true MedCPT Cross-Encoder baseline when available. A hard reranking subset is used for diagnosis: it contains test questions for which Hybrid top-100 includes at least one gold passage but Hybrid top-10 misses all gold passages. This subset is not a replacement for the full held-out test evaluation.

5.4 Validation Selection

Table 2 reports the enhanced candidate-generation validation selection. The selected w122 setting uses BM25 and dense retrieval as base sources, fielded BM25 as a stronger recall source, and MedCPT dual-encoder retrieval as an additional biomedical semantic source. It is selected because it maximizes validation Recall@100, matching the reranking objective of improving the candidate ceiling.

The complete candidate-fusion search space contains the original Hybrid setting $(1.0, 1.0, 0.0, 0.0)$ for BM25, dense, fielded BM25, and MedCPT dual encoder weights, plus w052 $(1.0, 1.0, 0.5, 0.2)$, w084 $(1.0, 1.0, 0.8, 0.4)$, w102 $(1.0, 1.0, 1.0, 0.2)$, and w122 $(1.0, 1.0, 1.2, 0.2)$. Only validation Recall@100 is used to choose among these settings; the held-out test split is evaluated after this selection. The LambdaMART hyperparameter search space is specified in Section 4, and model selection is likewise validation-only.

5.5 Downstream Generation Pilot

To connect evidence selection with the medical RAG setting, we run a controlled PubMedQA pilot with DashScope's OpenAI-compatible Qwen3-8B endpoint. The evaluation uses 100 PubMedQA questions from the common subset covered by Hybrid RRF, the available semantic cross-encoder PubMedQA output, and the KCH-style hypergraph reranker output. Each condition uses the same top-5 evidence passages, prompt template, temperature 0, maximum 256 output tokens, and no fine-tuning. The model must return one of `yes`, `no`, or `maybe` with passage citations. We report Accuracy, Macro-F1, citation support rate, unsupported claim rate, and answer-evidence entity consistency. This is a pilot diagnostic, not a replacement for the BioASQ retrieval evaluation.

Table 3: Held-out BioASQ evidence retrieval results for the enhanced candidate pool.

Method	Recall@10	MRR@10	nDCG@10	Recall@100
Original Hybrid RRF	0.4636	0.7550	0.5848	0.6077
Enhanced Hybrid w122	0.4660	0.7530	0.5844	0.7388
MedCPT Cross-Encoder	0.5172	0.7775	0.6390	0.7388
Enhanced KCH-MedRank	0.5332	0.7882	0.6446	0.7388

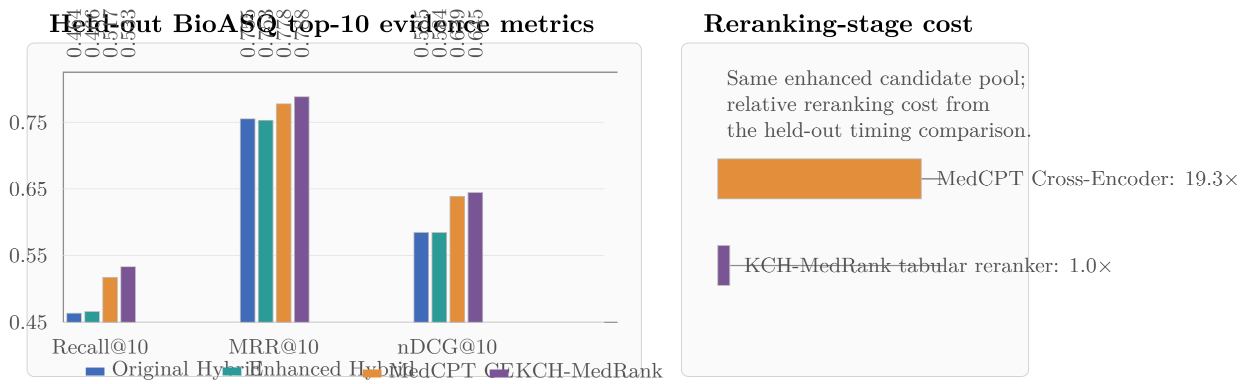

Figure 4: Main quantitative picture for the paper. KCH-MedRank improves held-out BioASQ top-10 evidence metrics over Hybrid RRF and MedCPT Cross-Encoder ordering, while the tabular reranking stage is much cheaper than full cross-encoder rescoring on the same candidate pool.

6 Results

6.1 Candidate Pool Quality

Table 3 summarizes the main held-out BioASQ retrieval results. The original Hybrid RRF test baseline has Recall@10 of 0.4636, MRR@10 of 0.7550, nDCG@10 of 0.5848, and Recall@100 of 0.6077. The enhanced w122 candidate pool was selected on validation Recall@100 and reaches test Recall@100 of 0.7388, providing a stronger candidate ceiling for reranking. This step is important because reranking cannot recover gold evidence that is absent from the candidate pool.

6.2 Main Held-Out Test Results

Enhanced KCH-MedRank reaches Recall@10 of 0.5332, MRR@10 of 0.7882, nDCG@10 of 0.6446, and Recall@100 of 0.7388. Relative to original Hybrid RRF, this corresponds to gains of +0.0696 Recall@10, +0.0332 MRR@10, and +0.0598 nDCG@10. Relative to the enhanced Hybrid RRF candidate ordering, KCH-MedRank improves Recall@10 by +0.0673, MRR@10 by +0.0352, nDCG@10 by +0.0602, and evidence coverage@10 by +0.0673.

Table 4 reports paired bootstrap testing against enhanced Hybrid RRF. All four top-10 metrics are significant with $p = 0.0002$. This confirms that the improvement is not only a consequence of the stronger candidate pool: the reranking step itself changes the top-10 evidence ordering in a measurable way.

Table 4: Paired bootstrap comparison between Enhanced KCH-MedRank and enhanced Hybrid RRF on the held-out BioASQ test split.

Metric	Hybrid RRF	KCH-MedRank	95% CI	p-value
Recall@10	0.4660	0.5332	[+0.0546, +0.0804]	0.0002
MRR@10	0.7530	0.7882	[+0.0224, +0.0481]	0.0002
nDCG@10	0.5844	0.6446	[+0.0509, +0.0701]	0.0002
Evidence coverage@10	0.4660	0.5332	[+0.0550, +0.0805]	0.0002

Table 5: Paired bootstrap comparison between Enhanced KCH-MedRank and MedCPT Cross-Encoder on the same enhanced candidate pool.

Metric	Baseline	KCH-MedRank	Delta	p-value
Recall@10	0.5172	0.5332	+0.0160	0.0074
MRR@10	0.7775	0.7882	+0.0107	0.1384
nDCG@10	0.6390	0.6446	+0.0056	0.2668

6.3 Comparison with Biomedical Semantic Reranking

On the same enhanced candidate pool, the true MedCPT Cross-Encoder baseline reaches Recall@10 of 0.5172, MRR@10 of 0.7775, and nDCG@10 of 0.6390. Table 5 reports paired bootstrap results against this strong semantic baseline. KCH-MedRank significantly improves Recall@10 ($p = 0.0074$), while MRR@10 and nDCG@10 are positive but not statistically significant.

The comparison with MedCPT Cross-Encoder is the stricter test for the proposed reranker. It shows that KCH-MedRank improves evidence coverage at top 10 beyond a strong biomedical semantic reranker, but the rank-sensitive MRR and nDCG gains should be described as positive trends rather than significant improvements. These results support the paper's evidence-retrieval claim. They do not yet justify making answer generation the primary contribution.

6.4 Hard Reranking and PubMedQA Diagnostics

Table 6 reports the diagnostic hard subset in which the enhanced Hybrid candidate pool contains gold evidence in the top 100 but misses all gold evidence in the top 10. This subset contains 65 held-out BioASQ questions and is not used as a replacement for the full test set. On this subset, enhanced Hybrid RRF has Recall@10 of 0 by construction, while full KCH-MedRank reaches Recall@10 of 0.4751. The result indicates that the reranker can recover a substantial fraction of missed top-10 evidence when the candidate pool contains recoverable gold passages. The subset also confirms the broader ablation pattern: retrieval and semantic features explain much of the recovery, and hypergraph/medical features should be interpreted as complementary rather than as a standalone source of all gains.

Table 7 gives a secondary PubMedQA retrieval diagnostic. PubMedQA has a different evidence structure and much stronger first-rank retrieval behavior than BioASQ, so it is used only as a robustness check. The lightweight HGB knowledge reranker improves Recall@10 from 0.8200 to 0.8953 over its Hybrid RRF source, with paired bootstrap Recall@10 delta of +0.0754 and $p = 0.0002$ in the experiment logs. MRR@10 changes only from 0.9850 to 0.9861 and is not significant, consistent with the observation that PubMedQA already has very high first-hit retrieval. This supports an evidence-coverage interpretation rather than a claim of broad answer-generation improvement.

Table 6: Diagnostic hard reranking subset on held-out BioASQ questions where enhanced Hybrid top-100 contains gold evidence but enhanced Hybrid top-10 misses it.

Method	Recall@10	MRR@10	nDCG@10	Recall@100
Enhanced Hybrid RRF	0.0000	0.0000	0.0000	0.8375
Retrieval-feature-only LambdaMART	0.4411	0.2180	0.2424	0.8375
LambdaMART + semantic, no hypergraph	0.4550	0.2821	0.2907	0.8375
Pairwise graph LTR	0.4415	0.2755	0.2768	0.8375
Full KCH-MedRank	0.4751	0.2742	0.2896	0.8375

Table 7: Secondary PubMedQA retrieval diagnostic on the held-out test split. The HGB knowledge reranker is a lightweight robustness reranker over the Hybrid candidate pool, not the main enhanced BioASQ KCH-MedRank configuration.

Method	Recall@10	MRR@10	nDCG@10	Recall@100
BM25	0.7284	0.9510	0.7490	0.8526
Dense	0.8644	0.9863	0.8760	0.9425
Hybrid RRF	0.8200	0.9850	0.8369	0.9350
Cross-encoder rerank	0.8268	0.9806	0.8381	0.9410
HGB knowledge reranker	0.8953	0.9861	0.9021	0.9350

6.5 Answer-Support Analysis

PubMedQA answer-selection analysis further separates evidence retrieval from answer correctness. For Hybrid and HGB evidence at top 10, the rule-based citation support rate is 1.0000 because at least one gold evidence passage is retrieved for each evaluated question. However, answer-supported rate remains limited by the lightweight yes/no/maybe answer selector: the majority baseline reaches 0.5550 supported answers, while TF-IDF logistic regression reaches 0.5050 for Hybrid and 0.4800 for HGB at top 10. Among the 98 questions with extracted question entities, answer-evidence entity consistency is 0.5510 at top 10. These analyses use qrels, gold labels, and deterministic entity overlap rather than an LLM judge. They support keeping generation secondary until stronger answer and faithfulness modeling is added.

Table 8 reports the controlled Qwen3-8B pilot. On 100 PubMedQA questions, KCH-style hypergraph evidence improves answer accuracy from 0.3400 to 0.3700 and Macro-F1 from 0.3506 to 0.3760 relative to Hybrid RRF evidence under the same prompt and decoding settings. Citation support is 0.9900 for all three evidence sources, so the pilot does not show a citation-support difference; it mainly suggests that evidence ordering can affect downstream label selection. Because the PubMedQA cross-encoder condition uses the available semantic cross-encoder output rather than a completed MedCPT PubMedQA rerun, this result is reported as a diagnostic RAG loop rather than a full secondary-dataset claim.

7 Ablation Study

The ablation study separates retrieval-strength gains from biomedical semantic, medical-knowledge, and hypergraph contributions. The evaluated variants remove the biomedical semantic score, MeSH hierarchy features, biomedical entity features, hypergraph diffusion and centrality, and PrimeKG relation features. All variants use the same enhanced top-100 candidate pool, so differences reflect reranking behavior rather than candidate availability.

Table 8: PubMedQA Qwen3-8B downstream generation pilot on 100 questions. All methods use the same prompt, top-5 evidence passages, temperature 0, and no fine-tuning. The PubMedQA cross-encoder condition uses the available semantic cross-encoder output, not a completed MedCPT PubMedQA run.

Evidence source + LLM	Accuracy	Macro-F1	Citation support	Unsupported claim	Entity consistency
Hybrid RRF + Qwen3-8B	0.3400	0.3506	0.9900	0.0100	0.5594
Semantic cross-encoder + Qwen3-8B	0.3400	0.3504	0.9900	0.0100	0.5782
KCH-style hypergraph reranker + Qwen3-8B	0.3700	0.3760	0.9900	0.0100	0.5751

Table 9: Ablation summary on the enhanced BioASQ held-out test split. Deltas are relative to enhanced Hybrid RRF.

Method	Recall@10	MRR@10	nDCG@10	ΔR@10
Enhanced Hybrid RRF	0.4660	0.7530	0.5844	–
Retrieval-feature-only LambdaMART	0.5208	0.7791	0.6282	+0.0548
LambdaMART + semantic, no hypergraph	0.5328	0.7885	0.6451	+0.0669
Pairwise graph LTR	0.5292	0.7876	0.6412	+0.0632
Hypergraph LTR without medical knowledge	0.5266	0.7782	0.6362	+0.0606
Full KCH-MedRank	0.5332	0.7882	0.6446	+0.0673
Remove semantic score	0.5244	0.7790	0.6330	+0.0584
Remove MeSH hierarchy	0.5355	0.7854	0.6441	+0.0696
Remove entity features	0.5318	0.7893	0.6443	+0.0658
Remove hypergraph diffusion/centrality	0.5341	0.7872	0.6452	+0.0681
Remove PrimeKG relations	0.5252	0.7834	0.6375	+0.0592

The interpretation is deliberately conservative. Biomedical semantic and retrieval features are strong. MeSH hierarchy and shared concept clusters provide medically interpretable features, but some leave-one-out ablations have small or mixed differences. PrimeKG exact-name relation features have low coverage and should remain auxiliary unless future normalization improves their contribution.

The ablation table suggests four points. First, learning-to-rank over retrieval features already provides a strong improvement over enhanced Hybrid RRF, raising Recall@10 from 0.4660 to 0.5208. This means the supervised query-group ranking objective itself is an important part of the method. Second, adding biomedical semantic evidence improves the strongest reranking variants: the semantic-without-hypergraph model reaches Recall@10 of 0.5328 and MRR@10 of 0.7885.

Third, graph structure and medical constraints are best interpreted as complementary features. Pairwise graph LTR and full KCH-MedRank are close, while the no-medical-knowledge hypergraph variant is weaker than the full model on Recall@10 and MRR@10. This supports the use of medical constraints but does not justify claiming that hyperedges alone dominate the improvement. Fourth, PrimeKG is not a primary driver in this implementation. Removing PrimeKG relation features still leaves Recall@10 of 0.5252 and MRR@10 of 0.7834, consistent with the observed sparsity of exact-name relation matching.

Table 10 adds paired bootstrap tests against the strongest ablations. Full KCH-MedRank is significantly better than Pairwise graph LTR on Recall@10 and nDCG@10, but its differences against LambdaMART + semantic without hypergraph, Remove MeSH hierarchy, and Remove hypergraph diffusion/centrality are not statistically significant. This result narrows the supported claim: the local hypergraph and medical-knowledge features improve interpretability and contribute useful signals, but the present evidence does not prove that any single feature group dominates the

Table 10: Paired bootstrap tests comparing Full KCH-MedRank with strong ablation variants on the enhanced BioASQ held-out test split. Deltas are Full minus ablation; brackets report 95% confidence intervals.

Baseline ablation	ΔMRR@10	ΔRecall@10	ΔnDCG@10
LambdaMART + semantic, no hypergraph	-0.0003 [-0.0047, 0.0040], $p = 0.8947$	0.0004 [-0.0032, 0.0041], $p = 0.8253$	-0.0005 [-0.0030, 0.0019], $p = 0.6895$
Pairwise graph LTR	0.0006 [-0.0043, 0.0056], $p = 0.7991$	0.0040 [0.0010, 0.0079], $p = 0.0054$	0.0035 [0.0011, 0.0062], $p = 0.0048$
Remove MeSH hierarchy	0.0028 [-0.0024, 0.0079], $p = 0.2938$	-0.0023 [-0.0075, 0.0026], $p = 0.3562$	0.0006 [-0.0027, 0.0037], $p = 0.7407$
Remove hypergraph diffusion/centrality	0.0010 [-0.0040, 0.0059], $p = 0.6855$	-0.0009 [-0.0049, 0.0026], $p = 0.6777$	-0.0006 [-0.0030, 0.0018], $p = 0.6421$

Table 11: Interpretation of close strong-ablation comparisons. The table clarifies what each comparison supports and what it does not support.

Comparison	Observed outcome	Supported interpretation
Full vs. LambdaMART + semantic, no hypergraph	No significant difference in MRR@10, Recall@10, or nDCG@10	The supervised retrieval and biomedical semantic features are the strongest performance drivers; the full model should not be claimed to dominate this ablation.
Full vs. Pairwise graph LTR	Significant gains in Recall@10 and nDCG@10, but not MRR@10	Hypergraph structure can improve top-10 evidence coverage over a pairwise graph representation, while first-rank precision remains close.
Full vs. Remove MeSH hierarchy	No significant difference; the ablation has slightly higher Recall@10	MeSH hierarchy features are best presented as interpretable medical constraints, not as an isolated source of statistically proven aggregate gain.
Full vs. Remove hypergraph diffusion/centrality	No significant difference; the ablation has slightly higher Recall@10 and nDCG@10	Diffusion and centrality features aid transparency and case-level explanation, but the current test does not prove that they independently increase overall ranking metrics.

ranking gain.

Taken together, the ablations support a combined framework: high-recall candidate generation, biomedical semantic reranking features, supervised listwise learning, and interpretable local hypergraph knowledge features together improve evidence retrieval.

8 Case Study

The failure analysis compares Enhanced Hybrid w122 with Enhanced KCH-MedRank at top 10 over 943 paired held-out BioASQ questions. KCH-MedRank rescues 661 gold passages into top 10 and loses 307 gold passages from top 10. This net gain supports the retrieval claim, but the lost cases are important because they show where knowledge-constrained features can overrule a strong lexical or semantic rank.

Figure 5 and Table 13 illustrate one rescued case and one lost case. In the rescued p-crAssphage example, the gold passage moves from rank 54 to rank 1. The main alignment signal is the shared bacteriophage concept, supported by a local cluster of phage-related candidates. This pattern is consistent with the method design: MeSH and local hypergraph features can promote evidence that is medically aligned even when the initial enhanced hybrid ordering places it outside the top 10.

The lost ubiquitination case shows the opposite behavior. A gold passage drops from rank 2 to rank 18 even though it discusses mono-ubiquitinated LDH-A and poly-ubiquitinated LDH. The question and passage contain related biochemical terminology, but the synonym and hierarchy

Table 12: Top-10 failure analysis comparing Enhanced Hybrid w122 with Enhanced KCH-MedRank on 943 paired held-out questions.

Diagnostic	Value
Paired questions	943
Questions with rescued gold evidence	414
Rescued gold passages	661
Questions with lost gold evidence	214
Lost gold passages	307
Net rescued passages	+354

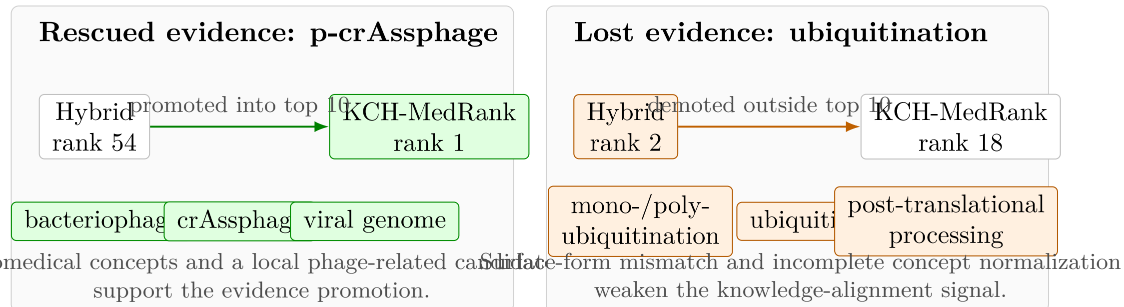

Figure 5: Visual case study of one rescued and one lost BioASQ evidence passage. The figure summarizes how local biomedical concept alignment can promote a missed gold passage, while incomplete synonym and hierarchy alignment can also demote a relevant passage.

features do not fully align the surface forms and broader post-translational-processing concepts. Similar losses occur for thyroid hormone analogs, cilengitide receptors, dasatinib treatment, GIST treatment, and Fanconi anemia. These examples motivate future improvements in biomedical concept normalization rather than stronger claims about the relation features.

9 Discussion

The evidence supports a focused claim: KCH-MedRank provides an interpretable medical evidence control layer that improves biomedical evidence retrieval over strong retrieval and semantic reranking baselines. The strongest statistically supported comparison against the true MedCPT Cross-Encoder baseline is Recall@10; MRR@10 and nDCG@10 move in the right direction but are not significant. Therefore, the safest interpretation is that KCH-MedRank is particularly useful for moving additional relevant evidence into the top-10 evidence set, rather than for guaranteeing better first-rank precision in every query.

The ablations suggest that the method should be understood as a combined biomedical reranking framework. Retrieval and semantic features remain strong contributors, while hypergraph diffusion, MeSH hierarchy, entity overlap, and relation features provide complementary signals. This is scientifically useful but also constrains the novelty claim: the contribution is not a standalone hypergraph formula, but a reproducible way to inject medical knowledge constraints into query-level reranking.

Additional selection diagnostics support keeping this interpretation conservative. A supplemental run that selected the full KCH-MedRank model primarily by validation Recall@10 reached held-out Recall@10 of 0.5294, below the reported validation-MRR-selected main model at 0.5332.

Table 13: Representative rescued and lost evidence cases from the held-out BioASQ comparison between Enhanced Hybrid w122 and KCH-MedRank.

Case	Gold rank	Question, signals, and interpretation
Rescued	54 → 1	Question: What is the p-crAssphage? Signals include bacteriophages, crAssphage, viral genome, and phylogeny. The gold passage shares the bacteriophage concept with the question and is connected to a local cluster of phage-related candidates, allowing the reranker to recover evidence missed by the enhanced hybrid ordering.
Lost	2 → 18	Question: Is there any protein that undergoes both mono-ubiquitination and poly-ubiquitination? Signals include ubiquitin, ubiquitination, ubiquitins, and post-translational processing. The passage is relevant but uses related biochemical forms and broader processing concepts; synonym and hierarchy features do not fully align mono-/poly-ubiquitination with the passage evidence.

Therefore, the paper retains the original validation policy rather than retuning to the test objective after observing held-out results. This diagnostic also suggests that simply changing the selection priority is insufficient; future gains are more likely to come from better biomedical concept normalization and more stable feature-group selection.

The PubMedQA Qwen3-8B pilot adds a downstream RAG-system diagnostic. Under fixed evidence count, prompt, model, and decoding settings, KCH-style hypergraph evidence improves answer accuracy and Macro-F1 over Hybrid RRF evidence on 100 PubMedQA questions. The citation-support rate is already high for all methods, so the pilot should not be read as proof of better faithfulness. It does show that white-box evidence selection can influence a downstream medical RAG answer loop without training or fine-tuning a large model. This strengthens the motivation for treating KCH-MedRank as an evidence control layer, while leaving full-scale generation evaluation and a completed PubMedQA MedCPT reranking condition as future work.

Several limitations remain. First, the method depends on a strong candidate pool; if top-100 recall is low, reranking cannot recover missing evidence. Second, exact-name PrimeKG mapping is sparse and should not be overclaimed. Third, strong-ablation bootstrap tests show that some differences between full KCH-MedRank and close ablations are not significant, so hypergraph, MeSH hierarchy, entity, and relation signals should be described as complementary and interpretable rather than dominant in isolation. Fourth, the Qwen3-8B generation result is a 100-question PubMedQA pilot; it should be scaled and repeated with a completed biomedical cross-encoder condition before being used as a central answer-generation claim.

9.1 Computational Profile

Table 14 summarizes the computational profile available from the experiment logs. The local hypergraph is built only over the top-M candidates for each question rather than over a global biomedical corpus, so graph construction and diffusion scale with the candidate pool and local feature coverage. In the enhanced BioASQ setting, the final learning-to-rank matrix has 45 features and 377,600 train-plus-validation rows, which keeps reranking within a conventional tabular ranking regime. Dense biomedical retrieval benefits from GPU acceleration. Cross-encoder rescoring is substantially more expensive: an early CPU run with a generic cross-encoder timed out after 2,400 seconds, while the final MedCPT Cross-Encoder baseline required a separate controlled full test run.

Table 14: Computational profile of the enhanced BioASQ experiment. The logs record scale, feature dimensionality, coverage, and hardware notes; full end-to-end wall-clock time is not available for every component.

Component	Recorded scale or setting	Implication
Processed data	4,719 questions; 40,221 passages; 42,608 qrels	Suitable for reproducible offline reranking.
Candidate pool	top-100 per question; 471,900 prediction rows	Reranking is bounded by local candidate pools rather than a global corpus graph.
Enhanced KCH-MedRank split	2,832 train, 944 validation, and 943 held-out test questions	Hyperparameters are selected before held-out testing.
Feature matrix	45 features; 283,200 training rows; 377,600 final train+validation rows	Learning-to-rank remains a tabular query-group problem.
Local hypergraph scope	question, top-100 passages, document, entity, MeSH, hierarchy, and optional relation nodes	Propagation cost scales with top-M and local feature coverage.
MeSH coverage	31,110 descriptors; 37,356/40,221 passages and 4,264/4,719 questions with MeSH matches	Controlled-vocabulary features are broadly available.
PrimeKG filtering	8,100,498 rows scanned; 5,766 local relations retained	Relation features are sparse and remain auxiliary.
Dense biomedical retrieval	MedCPT dual encoder used CUDA on an RTX 4060 Laptop GPU	GPU acceleration is practical for dense retrieval.
Cross-encoder scoring	Early generic CPU run timed out after 2,400 seconds; final MedCPT Cross-Encoder test baseline scored 943 questions and 94,300 candidates	Full cross-encoder rescoring is substantially more expensive than tabular reranking.

Table 15 reports a controlled reranking-stage timing comparison on the held-out BioASQ test split. Both methods use the same enhanced top-100 candidate pool over 943 questions and 94,300 candidate passages. The comparison excludes first-stage candidate generation and offline LambdaMART training. Under this setting, MedCPT Cross-Encoder scoring on CUDA takes 699.46 seconds, while KCH-MedRank test-time feature construction plus LightGBM scoring on CPU takes 36.31 seconds. Thus KCH-MedRank is 19.3× faster at the reranking stage while reaching higher Recall@10, MRR@10, and nDCG@10 on the same candidate pool. This strengthens the practical interpretation of the main result: the observed Recall@10 gain over MedCPT Cross-Encoder is small but meaningful because it is obtained with lower reranking latency and inspectable features. End-to-end production benchmarking, including first-stage retrieval and data-loading overhead, remains future work.

Future work should improve biomedical concept normalization, add better synonym and abbreviation handling, and scale controlled answer generation after evidence support metrics are complete. A stronger generation study should measure citation support, unsupported claims, and answer-evidence entity consistency, not only answer accuracy.

10 Conclusion

This paper presented KCH-MedRank, an interpretable medical evidence control layer implemented as a lightweight and reproducible knowledge-constrained local hypergraph learning-to-rank frame-

Table 15: Reranking-stage efficiency on the held-out BioASQ test split using the same enhanced top-100 candidate pool. Timing excludes first-stage candidate generation and offline LambdaMART training; KCH-MedRank includes test-time feature construction and LightGBM scoring, while MedCPT Cross-Encoder includes tokenization and forward scoring.

Method	Seconds	ms/query	Cand./s	Recall@10	MRR@10	nDCG@10	Notes
MedCPT Cross-Encoder	699.46	741.74	134.82	0.5172	0.7775	0.6390	tokenization + forward
KCH-MedRank	36.31	38.50	2597.07	0.5332	0.7882	0.6446	features + LightGBM

work. The method keeps first-stage retrieval practical, constructs a local cross-granularity hypergraph over question, passage, document, entity, MeSH, and relation signals, and uses the resulting medical and structural signals as interpretable learning-to-rank features.

The BioASQ results show stronger top-10 evidence retrieval than original and enhanced Hybrid RRF, with statistically significant improvements across top-10 retrieval metrics against enhanced Hybrid RRF. Against a true MedCPT Cross-Encoder baseline on the same enhanced candidate pool, the strongest supported gain is statistically significant Recall@10 improvement; MRR@10 and nDCG@10 remain positive but non-significant trends. A controlled reranking-stage timing comparison further shows that KCH-MedRank is 19.3× faster than MedCPT Cross-Encoder on the held-out test pool when candidate generation and offline training are excluded. Strong-ablation tests show that graph and knowledge signals are complementary to retrieval and semantic learning-to-rank features rather than dominant in isolation.

The PubMedQA Qwen3-8B pilot indicates that evidence ordering can affect downstream answer selection under a fixed prompt and decoding setup, improving Accuracy and Macro-F1 for the KCH-style evidence condition while citation support remains high across all compared evidence sources. Overall, the results support KCH-MedRank as a white-box evidence selection and grounding module for evidence-grounded medical QA. The method should not be viewed as a clinically validated answer generation or decision-support system.

References

[1] Payal Chandak, Kexin Huang, and Marinka Zitnik. Building a knowledge graph to enable precision medicine. *Scientific Data*, 10:67, 2023. doi: 10.1038/s41597-023-01960-3. URL `https://www.nature.com/articles/s41597-023-01960-3`. Accessed 2026-05-21.

[2] Gordon V. Cormack, Charles L. A. Clarke, and Stefan Buettcher. Reciprocal rank fusion outperforms condorcet and individual rank learning methods. In *Proceedings of the 32nd International ACM SIGIR Conference on Research and Development in Information Retrieval*, pages 758–759. ACM, 2009. doi: 10.1145/1571941.1572114. URL `https://research.google/pubs/reciprocal-rank-fusion-outperforms-condorcet-and-individual-rank-learning-methods/`. Accessed 2026-05-21.

[3] Darren Edge, Ha Trinh, Newman Cheng, Joshua Bradley, Alex Chao, Apurva Mody, Steven Truitt, and Jonathan Larson. From local to global: A graph RAG approach to query-focused summarization. *arXiv preprint arXiv:2404.16130*, 2024. doi: 10.48550/arXiv.2404.16130. URL `https://arxiv.org/abs/2404.16130`. Accessed 2026-05-21.

[4] Qiao Jin, Bhuwan Dhingra, Zhengping Liu, William Cohen, and Xinghua Lu. PubMedQA: A dataset for biomedical research question answering. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International*

Joint Conference on Natural Language Processing (EMNLP-IJCNLP), pages 2567–2577. Association for Computational Linguistics, 2019. doi: 10.18653/v1/D19-1259. URL https://aclanthology.org/D19-1259/. Accessed 2026-05-21.

[5] Qiao Jin, Won Kim, Qingyu Chen, Donald C. Comeau, Lana Yeganova, W. John Wilbur, and Zhiyong Lu. MedCPT: Contrastive pre-trained transformers with large-scale PubMed search logs for zero-shot biomedical information retrieval. *Bioinformatics*, 39(11):btad651, 2023. doi: 10.1093/bioinformatics/btad651. URL https://academic.oup.com/bioinformatics/article/39/11/btad651/7335842. Accessed 2026-05-21.

[6] Guolin Ke, Qi Meng, Thomas Finley, Taifeng Wang, Wei Chen, Weidong Ma, Qiwei Ye, and Tie-Yan Liu. LightGBM: A highly efficient gradient boosting decision tree. In *Advances in Neural Information Processing Systems 30*, 2017. URL https://proceedings.neurips.cc/paper/2017/hash/6449f44a102fde848669bdd9eb6b76fa-Abstract.html. Accessed 2026-05-21.

[7] Anastasia Krithara, Anastasios Nentidis, Konstantinos Bougiatiotis, and Georgios Paliouras. BioASQ-QA: A manually curated corpus for biomedical question answering. *Scientific Data*, 10(1):170, 2023. doi: 10.1038/s41597-023-02068-4. URL https://pubmed.ncbi.nlm.nih.gov/36973320/. Accessed 2026-05-21.

[8] Patrick Lewis, Ethan Perez, Aleksandra Piktus, Fabio Petroni, Vladimir Karpukhin, Naman Goyal, Heinrich Kuttler, Mike Lewis, Wen-tau Yih, Tim Rocktaschel, Sebastian Riedel, and Douwe Kiela. Retrieval-augmented generation for knowledge-intensive NLP tasks. *arXiv preprint arXiv:2005.11401*, 2020. doi: 10.48550/arXiv.2005.11401. URL https://arxiv.org/abs/2005.11401. Accessed 2026-05-21.

[9] Haoran Luo, Haihong E, Guanting Chen, Yandan Zheng, Xiaobao Wu, Yikai Guo, Qika Lin, Yu Feng, Zemin Kuang, Meina Song, Yifan Zhu, and Luu Anh Tuan. HyperGraphRAG: Retrieval-augmented generation via hypergraph-structured knowledge representation. *arXiv preprint arXiv:2503.21322*, 2025. doi: 10.48550/arXiv.2503.21322. URL https://arxiv.org/abs/2503.21322. Accessed 2026-05-21.

[10] National Library of Medicine. Medical subject headings (MeSH). https://www.nlm.nih.gov/mesh/meshhome.html, 2026. URL https://www.nlm.nih.gov/mesh/meshhome.html. Accessed 2026-05-21.

[11] Stephen E. Robertson and Hugo Zaragoza. The probabilistic relevance framework: BM25 and beyond. *Foundations and Trends in Information Retrieval*, 3(4):333–389, 2009. doi: 10.1561/1500000019. URL https://ir.webis.de/anthology/2009.ftir_journal-ir0anthology0volumeA3A4.0/. Accessed 2026-05-21.

[12] George Tsatsaronis, Georgios Balikas, Prodromos Malakasiotis, Ioannis Partalas, Matthias Zschunke, Michael R. Alvers, Dirk Weissenborn, Anastasia Krithara, Sergios Petridis, Dimitris Polychronopoulos, Yannis Almirantis, John Pavlopoulos, Nicolas Baskiotis, Patrick Gallinari, Thierry Artieres, Axel-Cyrille Ngonga Ngomo, Norman Heino, Eric Gaussier, Laura Barrio-Alvers, Michael Schroeder, Ion Androutsopoulos, and Georgios Paliouras. An overview of the BIOASQ large-scale biomedical semantic indexing and question answering competition. *BMC Bioinformatics*, 16:138, 2015. doi: 10.1186/s12859-015-0564-6. URL https://link.springer.com/article/10.1186/s12859-015-0564-6. Accessed 2026-05-21.

[13] Changjian Wang, Weihong Deng, Weili Guan, Quan Lu, and Ning Jiang. Cross-granularity hypergraph retrieval-augmented generation for multi-hop question answering. In *Proceedings of the AAAI Conference on Artificial Intelligence*, 2026. URL `https://ojs.aaai.org/index.php/AAAI/article/download/40623/44584`. Accessed 2026-05-21.

[14] Junde Wu, Jiayuan Zhu, Yunli Qi, Jingkun Chen, Min Xu, Filippo Menolascina, and Vicente Grau. Medical graph RAG: Towards safe medical large language model via graph retrieval-augmented generation. *arXiv preprint arXiv:2408.04187*, 2024. doi: 10.48550/arXiv.2408.04187. URL `https://arxiv.org/abs/2408.04187`. Accessed 2026-05-21.

[15] Guangzhi Xiong, Qiao Jin, Zhiyong Lu, and Aidong Zhang. Benchmarking retrieval-augmented generation for medicine. In *Findings of the Association for Computational Linguistics: ACL 2024*, pages 6233–6251. Association for Computational Linguistics, 2024. doi: 10.18653/v1/2024.findings-acl.372. URL `https://aclanthology.org/2024.findings-acl.372/`. Accessed 2026-05-21.

[16] Guangzhi Xiong, Qiao Jin, Xiao Wang, Minjia Zhang, Zhiyong Lu, and Aidong Zhang. Improving retrieval-augmented generation in medicine with iterative follow-up questions. *arXiv preprint arXiv:2408.00727*, 2024. doi: 10.48550/arXiv.2408.00727. URL `https://arxiv.org/abs/2408.00727`. Accessed 2026-05-21.

[17] Fangfang Xu, Yifan Kuai, Chao Gao, Dan Luo, and Jinguang Gu. Medical HyperRAG: A hypergraph-enhanced retrieval-augmented generation framework for clinical question answering. *Research Square preprint*, 2026. doi: 10.21203/rs.3.rs-8497459/v1. URL `https://doi.org/10.21203/rs.3.rs-8497459/v1`. Accessed 2026-05-21.